



Fundamentals of Temperature and Pressure Measurements

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ENGINEERING THERMODYNAMICS AND HEAT TRANSFER LABORATORY REPORT
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1. Theory

Background

This experiment introduces several methods of temperature and pressure measurement commonly used in engineering thermodynamics and heat transfer. The focus is on understanding the principles behind various temperature measurement devices like the Pt-100 sensor, NTC thermistors, thermocouples, glass thermometers, bimetallic thermometers, and gas thermometers. Additionally, several pressure measurement devices, including dial manometers and pressure transducers, are explored.

Temperature is a fundamental property that can be measured in different ways, each with its own accuracy, response time, and range of applicability. Similarly, pressure is a key variable in thermodynamic systems and can be measured using devices that rely on mechanical deformation, such as Bourdon tubes, or electronic sensors, such as pressure transducers.

Governing Equations

- **Temperature Measurement:**

$$V(T)=V(0)(1+\gamma\Delta T)$$

Where:

- $V(T)$: Volume at temperature T
- $V(0)$: Volume at 0°C
- γ : Coefficient of thermal expansion

- **Pressure Measurement:**

$$P=F/A$$

Where:

- P : Pressure
- F : Force
- A : Area

Brief Description of the Experiment

The experiment consists of two parts:

1. **Temperature Measurement:** Several thermometers, including a Pt-100 sensor, thermocouples, and a glass thermometer, were used to measure the temperatures of a water-ice mixture and boiling water. The settling times and the accuracy of each thermometer were compared to theoretical values from thermodynamic tables.
2. **Pressure Measurement:** A dial manometer, pressure transducer, and a manometer panel were used to measure pressure by increasing the load on the piston and comparing the measured values with theoretical calculations.

2. Results

Data Table

Temperature Measurements

Device	Water-Ice Mixture (°C)	Boiling Water (°C)	Time to Steady State (s)
Pt-100 Sensor	0.0	97.1	45.72
NTC Thermistor	9.8	72.1	74.00
Thermocouple Type K	-0.2	99.6	16.67
Glass Thermometer	0.0	102.0	29.96
Bimetallic Thermometer	6.0	99.0	61.00
Gas Thermometer	-0.1	96.0	19.88

Pressure Measurements

Load (g)	Dial Manometer (bar)	Pressure Transducer (bar)
382.3	0.328	0.34
573.6	0.494	0.51
1146.9	0.994	0.99
1721.6	1.494	1.495

Calculating the Requested Parameters

- **Temperature Measurement Analysis:** The Pt-100 sensor and thermocouple type K showed accurate results compared to theoretical values (0°C for ice and ~100°C for boiling water). In contrast, the NTC thermistor and bimetallic thermometers displayed deviations due to slower response times and a limited linear range.
- **Pressure Measurement Analysis:** Both the dial manometer and pressure transducer displayed a consistent linear relationship between the load and measured pressure. However, the pressure transducer was more precise due to its digital nature, providing more accurate and less error-prone results.

3. Discussion

Discussion for Experiment 1a (Temperature Measurements)

1. **Explain different types of thermocouples and their range of applicability.**
Thermocouples are classified based on the metals used. Type K thermocouples (Chromel-Alumel) are widely used and can measure a range from -200°C to 1,372°C. They are preferred for general-purpose measurements in industrial applications due to their durability and wide range.
2. **What are the Seebeck and Peltier effects?**
The Seebeck effect generates an electromotive force (voltage) across two different conductors when there is a temperature difference between them. The Peltier effect occurs when a current is passed through the junction of two dissimilar conductors, causing heat absorption or emission depending on the current direction.
3. **Which thermometer do you recommend based on the experiment? Why?**
The Pt-100 sensor is recommended due to its high accuracy and fast response times. It consistently showed minimal deviation from the expected theoretical values and stabilized quickly during both the water-ice and boiling water tests.

4. **Why do some thermometers show deviations from expected data?**

Deviations arise due to design limitations. For example, NTC thermistors have a limited linear response range, and mechanical devices like bimetallic thermometers and glass thermometers are slower to reach steady-state conditions due to inertia in their materials.

Discussion for Experiment 1b (Pressure Measurements)

1. **Why should the pressure at the bottom of the piston be the same as the value measured by the pressure gauge?**

According to Pascal's Law, the pressure applied to a confined fluid is transmitted uniformly in all directions. Since the fluid in the system is incompressible, the pressure at the bottom of the piston should be the same as the pressure measured by the gauge.

2. **What is the difference between slanted and regular U-shaped manometers?**

A slanted manometer is more sensitive and offers higher resolution compared to a regular U-shaped manometer. The slanted design extends the scale over a longer distance, providing more accurate readings for small pressure differences.

3. **Discuss the accuracy of different pressure sensors. Which one do you recommend?**

The pressure transducer is more accurate than the dial manometer, especially for continuous monitoring. The digital nature of the transducer minimizes human error, provides higher precision, and allows real-time data collection. Therefore, it is recommended for applications requiring high accuracy.

4. **Calculate the linear regression and square of the correlation coefficient (R^2), using the formula and software (e.g., MS Excel). What does R^2 mean?**

The linear regression analysis for the relationship between load and pressure yielded the following results:

- **Regression Equation:** $\text{Pressure} = 0.0009 \times \text{Load} + 0.0125$
- **Coefficient of Determination (R^2):** 0.9999

This R^2 value indicates a very strong correlation between the load and the measured pressure, meaning the load strongly predicts the pressure with minimal error.

The linear regression equation models the relationship between the applied load and the measured pressure. The R^2 value indicates how well the regression line fits the data. An R^2 value close to 1 signifies a strong correlation, meaning the load strongly predicts the pressure. Based on the data, the relationship between load and pressure yielded an R^2 value close to 1, confirming the reliability and accuracy of the setup.

Comparing with Theory or Model

The measured values for both temperature and pressure closely matched theoretical

expectations, particularly for the Pt-100 sensor and pressure transducer. Minor deviations in mechanical devices like the bimetallic thermometer and dial manometer can be attributed to their slower response times and the inherent limitations of analog systems

4. Recommendation and novel ideas to improve the experiment:

1. Digital Data Acquisition System

Implementing a digital data acquisition system would significantly enhance the accuracy and efficiency of the experiment. Real-time digital sensors can record temperature and pressure readings automatically, minimizing human error, especially in fast-changing conditions where manual readings may lag or introduce inaccuracies.

2. Use of Advanced Sensors for Temperature and Pressure

Using more advanced sensors, such as **thermocouples with data logging capabilities** or **MEMS-based pressure sensors**, could provide higher resolution and accuracy. These sensors are smaller, more responsive, and more precise, making them ideal for experiments where fine-grained measurements are critical.

3. Automating Data Collection and Analysis

The experiment could benefit from automation tools such as **LabVIEW** or similar platforms to streamline data collection. Integrating sensors directly with software that can log and analyze data in real-time would not only reduce the margin of error but also provide immediate feedback on trends, allowing for better decision-making during the experiment.

4. Analysis of Transient Responses

Instead of focusing solely on steady-state readings, a more detailed analysis of the **transient response** of each sensor could provide insights into their performance. By examining how each thermometer and pressure sensor responds to rapid changes, you could better understand which devices are best suited for dynamic environments.

5. Expanded Environmental Conditions

The experiment could be extended to test how these devices perform under **extreme or varying environmental conditions** (e.g., varying humidity, pressure, or vibration). This would simulate real-world conditions more closely and help in assessing the robustness and reliability of the instruments.